Fundamental Trigonometric Identities

ANSWER KEY

Section 1: Reciprocal and Quotient Identities

1 Simplify the expression $\tan x \cdot \csc x$.

a. $\sec x$ ✓

b. $\cos x$

c. $\sin x$

d. $\cot x$

$\tan x \cdot \csc x = \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \frac{1}{\cos x} = \sec x$.

2 Simplify $\frac{\cos x}{\cot x}$.

a. $\sin x$ ✓

b. $\cos^2 x$

c. $\tan x$

d. $\csc x$

$\frac{\cos x}{\cos x / \sin x} = \cos x \cdot \frac{\sin x}{\cos x} = \sin x$.

3 If $\sin\theta = -3/5$ and $\tan\theta > 0$, find $\cos\theta$.

a. $-4/5$ ✓

b. $4/5$

c. $-3/4$

d. $5/4$

$\sin < 0$ and $\tan > 0 \Rightarrow$ Quadrant III. In QIII, $\cos\theta = -\sqrt{1-(-3/5)^2} = -\sqrt{16/25} = -4/5$.

Section 2: Pythagorean Identities

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4 Which identity is directly derived by dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$?

a. $1 + \tan^2\theta = \sec^2\theta$ ✓

b. $1 + \cot^2\theta = \csc^2\theta$

c. $\sin^2\theta = 1 - \cos^2\theta$

d. $\tan^2\theta + \cot^2\theta = 1$

$\frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} \Rightarrow \tan^2\theta + 1 = \sec^2\theta$.

5 Simplify $(\sec\theta - 1)(\sec\theta + 1)$.

a. $\tan^2\theta$ ✓

b. $\cot^2\theta$

c. $\cos^2\theta$

d. 1

Difference of squares: $\sec^2\theta - 1 = \tan^2\theta$.

6 Simplify $\frac{1 - \cos^2 x}{\sin x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\csc x$

d. 1

$\frac{\sin^2 x}{\sin x} = \sin x$.

Section 3: Even/Odd Identities and Symmetry

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7 Which trigonometric functions are even functions ($f(-\theta) = f(\theta)$)?

a. $\cos\theta$ **and** $\sec\theta$ ✓

b. $\sin\theta$ and $\csc\theta$

c. $\tan\theta$ and $\cot\theta$

d. All six functions

Only cosine and its reciprocal secant are even: $\cos(-\theta) = \cos\theta$ and $\sec(-\theta) = \sec\theta$.

8 Simplify $\sin(-\theta) \cdot \cot(-\theta)$.

a. $\cos\theta$ ✓

b. $-\cos\theta$

c. $\sin\theta$

d. $-\sin\theta$

$\sin(-\theta) = -\sin\theta$ and $\cot(-\theta) = -\cot\theta$. Product $= (-\sin\theta)(-\cot\theta) = \sin\theta \cdot \frac{\cos\theta}{\sin\theta} = \cos\theta$.

9 Simplify $\frac{\tan(-x)}{\sin(-x)}$.

a. $\sec x$ ✓

b. $-\sec x$

c. $\cos x$

d. $-\cos x$

$\frac{-\tan x}{-\sin x} = \frac{\tan x}{\sin x} = \frac{\sin x/\cos x}{\sin x} = \frac{1}{\cos x} = \sec x$.

Section 4: Verifying Identities Algebraically

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10 Prove $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$. What technique verifies this algebraically?

a. **Multiply numerator and denominator of the left side by the conjugate** $(1-\cos x)$ ✓

b. Square both sides

c. Take the derivative

d. Cross multiply and set to 0

$\frac{\sin x(1-\cos x)}{(1+\cos x)(1-\cos x)} = \frac{\sin x(1-\cos x)}{1-\cos^2 x} = \frac{\sin x(1-\cos x)}{\sin^2 x} = \frac{1-\cos x}{\sin x}$.

11 Simplify $\frac{\sec x - \cos x}{\sin x}$.

a. $\tan x$ ✓

b. $\cot x$

c. $\sec x$

d. $\sin x$

$\frac{1/\cos x - \cos x}{\sin x} = \frac{(1-\cos^2 x)/\cos x}{\sin x} = \frac{\sin^2 x}{\cos x \sin x} = \frac{\sin x}{\cos x} = \tan x$.

12 Simplify $(1+\tan^2\theta)(1-\sin^2\theta)$.

a. 1 ✓

b. $\sec^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

$(1+\tan^2\theta) = \sec^2\theta$ and $(1-\sin^2\theta) = \cos^2\theta$. Product $= \sec^2\theta \cdot \cos^2\theta = 1$.

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